

Sofiène Zayati

Full-Stack & AI Engineer | Embedded Systems Background

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/ Tunis, Tunisia

PROFILE

Full-stack and AI engineer pursuing an Engineering Degree in Web and Internet Technologies at ESPRIT, backed by a Bachelor's Degree in Embedded and Mobile Systems. Develops web products, applied AI workflows, Python and Node.js services, and connected ESP32/WT32 systems across interfaces, APIs, data, deployment, and hardware.

ENGINEERING FOCUS

Product systems | Applied AI | API architecture | Automation | Embedded IoT | Delivery

PROFESSIONAL EXPERIENCE

Jul 2025 - Aug 2025

AI Automation & Full-Stack Developer

Inspark Connect | Tunis, Tunisia

- Developed n8n and Gemini assistants integrated with a Laravel, Vue.js, and MySQL e-commerce platform for product creation, enrichment, campaigns, sales reporting, and logistics analysis.
- Implemented guarded intent routing, JSON validation and sanitization, relational SQL aggregation, SMTP and REST integrations, and fallback handling for structured business outputs.

Feb 2024 - May 2024

Embedded Systems & Software Developer

Scheidt & Bachmann Maghreb | Graduation internship | Megrine, Tunisia

- Engineered MacroPark across five Scrum sprints, integrating administration, Flutter mobile access, Quercus LPR, FastAPI services, Mosquitto MQTT, and WT32-ETH01 barrier firmware.
- Validated camera/MQTT and mobile/BLE access paths, four relay channels, persistent controller configuration, OTA maintenance, access history, and Docker Compose deployment.

Jan 2023 - Feb 2023

Hardware & Software Developer Intern

Scheidt & Bachmann Maghreb | Megrine, Tunisia

- Prototyped Smart Unlock using ESP32 scanning, Eddystone BLE identifier checks, relay and LED actuation, and Mosquitto MQTT status messaging.

Jan 2022 - Feb 2022

Web Development Intern

Sotunol | Megrine, Tunisia

- Developed and improved the company's web platform across user-facing interfaces and backend performance.

EDUCATION

Sep 2024 - Present

Engineering Degree in Web and Internet Technologies (TWIN)

ESPRIT School of Engineering | Ariana, Tunisia

Sep 2021 - Jun 2024

Bachelor's Degree in Embedded and Mobile Systems

Higher Institute of Technological Studies of Rades | Tunisia

Sep 2017 - Jun 2021

Technical Sciences Baccalaureate

Lycee Technique Farhat Hached | Rades, Tunisia

SELECTED TECHNICAL PROJECTS

SmartProperty - AI-Powered Real Estate Platform2026

- Engineered four-role workflows with React, NestJS, FastAPI, MongoDB, Socket.IO, Stripe, Cloudinary, and OpenStreetMap across listings, discovery, visits, applications, payments, analytics, and administration.
- Integrated XGBoost valuation, recommendations, CLIP/BLIP image understanding, OCR-based solvency review, and Groq assistance with containerized delivery configuration.

sofienezayati.tn/project/smartproperty

Machetamache - Recipe & Food-Costing Product2026

- Delivered a React 19, Express 5, and SQLite workspace for a working chef, covering recipes, exact costing, portion scaling, print-ready kitchen sheets, backups, and PC/mobile LAN access.
- Implemented integer-based calculations, Vitest/Supertest verification, network-safety and legacy-phone fallbacks, and a portable Windows release.

sofienezayati.tn/project/machetamache

Prigado AI - E-Commerce Automation2025

- Built conversational n8n and Gemini workflows for product creation and enrichment, campaign generation, seller reporting, product trends, and logistics risk analysis.
- Connected Laravel, Vue.js, MySQL, SMTP, and REST services through guarded intent routing, sanitized JSON, and structured SQL aggregation.

sofienezayati.tn/project/prigado

MacroPark - Connected Parking Access2024

- Connected Quercus LPR and Flutter/BLE access to FastAPI, MQTT, and WT32-ETH01 firmware across administration, mobile access, services, and physical barrier control.
- Validated two access paths, four barrier channels, SPIFFS persistence, OTA updates, relay behavior, and containerized multi-service deployment across five Scrum sprints.

sofienezayati.tn/project/macropark

Additional work includes TounsiAds, InnoMall, Genius PMO HRMS, Green Coffee OS, Interna, and Smart Agri - an ESP32 crop-monitoring system with temperature, humidity, soil-moisture, Blynk telemetry, and local indicators. Full case studies: sofienezayati.tn.

TECHNICAL TOOLKIT

Programming	TypeScript, JavaScript, Python, Java, C, C++, SQL, PHP
Frontend	React, Next.js, Angular, Vue.js, Tailwind CSS, JavaFX
Backend & APIs	NestJS, Express, FastAPI, Spring Boot, Laravel, Symfony, REST, WebSockets
AI & Automation	n8n, Gemini, Groq, XGBoost, CLIP/BLIP, OCR, recommendation systems
Data	MongoDB, MySQL, PostgreSQL, SQLite, Firebase
Embedded & IoT	ESP32, WT32-ETH01, Arduino, MQTT, BLE, Ethernet/Wi-Fi, sensors, relays, OTA
Delivery	Git, Linux, Docker, Docker Compose, Kubernetes, Jenkins, SonarQube, CI/CD
Engineering	API design, realtime systems, testing, responsive UX, hardware-software integration

LANGUAGES

Arabic Native	English Fluent	French Fluent	German Goethe-Zertifikat A2
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ENGINEERING RANGE

Web products, applied AI, and connected devices. Experience spans product interfaces, APIs, data models, automation, realtime messaging, containerized services, embedded firmware, and physical control.